



THE UNIVERSITY *of* EDINBURGH
School of Mathematics

Financial Mathematics **MATH10003**

Wednesday, 11th May 2022

1300-1500 † *

† **All students:** You have an additional **1 hour** to assemble and submit your PDF.

Final submission deadline: 16:00

* **Students with a Schedule of Adjustment for additional time in examinations:**
You are entitled to a further fixed additional **1 hour** for this remote examination.

Final submission deadline: 17:00

Attempt all questions

Important instructions

1. Start each question on a new sheet of paper.
2. Number your sheets of paper to help you scan them in order.
3. Only write on one side of each piece of paper.
4. If you have rough work to do, simply include it within your overall answer – put brackets at the start and end of it if you want to highlight that it is rough work.

- (1) Assume that today's stock price, S , is £40 and it is expected either to increase by 25% or to decrease by 20% every four months over the next two four-month periods (two-period Binomial tree, $n = 0, 1, 2$). Also, suppose that the continuously compounded risk-free rate of interest is 6% per annum and that S does not pay dividends.

- (a) Verify the absence of arbitrage and compute the prices (at period $n = 0$ and $n = 1$, i.e. $t = 0$ and $t = 4$ months) of the 8-month European Call option Φ with strike price $K = £41$, written on the stock S .

[13 marks]

- (b) Compute the prices (at $n = 0$ and $n = 1$) of the 8-month American call option Φ with strike price $K = £41$, written on the stock S . Is an early exercise optimal?

[9 marks]

- (c) Recall that the binomial tree aims at being an approximation to the real evolution of the price of a stock. Does this 2-period binomial model reproduce a specific volatility σ of the real observed market price of a stock? Explain.

[3 marks]

- (d) In relation to the 2-period binomial model described above, compute the fair delivery price K at time $t = 0$ and $t = \text{"four-months"}$ of a forward contract with maturity at *eight-months* and written on the underlying S .

[3 marks]

- (e) A trader wants to buy a power option with maturity four months and payoff $\Phi^2(x) = \max\{K - x^2, 0\}$, where $K = 40^2$, but cannot find anyone willing to sell it. Find a replicating strategy for this option and determine the amount of initial capital required.

Verify that your strategy replicates Φ^2 . What would be a fair price for the option Φ^2 ?

[11 marks]

- (2) Let $(W_t^1)_{t \geq 0}$ and $(W_t^2)_{t \geq 0}$ be two independent Brownian motions.

- (i) Define what a Brownian motion is.

[4 marks]

- (ii) Verify whether the following processes are Brownian motions.

(a) $\frac{1}{2}(W_t^1 + W_t^2)$

[5 marks]

(b) $\frac{1}{\sqrt{2}}(W_t^1 - W_t^2)$

[12 marks]

(c) $\frac{1}{2}W_{4t}^1$

[12 marks]

(3) Let $T \in [0, \infty)$, W be a Brownian motion and $X, Y \in L^2([0, T])$.

(a) Using Itô's isometry and $2ab = (a + b)^2 - a^2 - b^2$, show that

$$\mathbb{E} \left[\int_0^T X_s dW_s \int_0^T Y_s dW_s \right] = \mathbb{E} \left[\int_0^T X_s Y_s ds \right].$$

[7 marks]

(b) Calculate the covariance of A and B , where $A = \int_0^T (T - s) dW_s$ and $B = \int_0^T s dW_s$.

[4 marks]

(4) Assume $T > 0$ and $b \in \mathbb{R}$, $a, r, \sigma > 0$ being constants. Let $(X_s)_{s \in [0, T]}$ be the Itô process defined as

$$\begin{aligned} dX_s &= (b - aX_s)ds + \sigma dW_s, \quad \text{for } s \in [0, T] \\ X_0 &= x. \end{aligned}$$

Let $f \in C^{1,2}([0, T] \times \mathbb{R}, \mathbb{R})$ and assume that f fulfils for $(t, x) \in [0, T] \times \mathbb{R}$ the equation

$$\partial_t f(t, x) + (b - ax)\partial_x f(t, x) + \frac{1}{2}\sigma^2 \partial_{xx} f(t, x) - rf(t, x) = 0, \quad (1)$$

and also the boundary condition $f(T, x) = \Phi(x)$ for $x \in \mathbb{R}$ for some function $\Phi \in C^1(\mathbb{R}, \mathbb{R})$.

(a) Apply Itô's formula to $e^{-rs}f(s, X_s)$ to obtain an expression of the form $de^{-rs}f(s, X_s) = \dots ds + \dots dW_s$.

[4 marks]

(b) Use Equation (1) to verify that

$$e^{-rt}f(t, X_t) = e^{-rT}f(T, X_T) - \int_t^T e^{-rs}\sigma(s, X_s)\partial_x f(s, X_s)dW_s.$$

[6 marks]

(c) Assume that the process $\left(e^{-rs}\sigma(s, X_s)\partial_x f(s, X_s)\right)_{s \in [0, T]} \in L^2([0, T])$. Prove that for any $x \in \mathbb{R}$ the value $f(0, x)$ can be expressed by

$$f(0, x) = e^{-r(T)}\mathbb{E}[\Phi(X_T)].$$

[7 marks]